



CS182: Introduction to Machine Learning – Decision Trees

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Q & A:

How do these in-class polls work?

- Scan QR Code for accessing the polls
- Answer all poll questions **during lecture for full credit** or **within 24 hours for half credit**
- You have 8 free “poll points” for the semester that will excuse you from all polls from a single lecture; you cannot use more than 3 poll points consecutively.

Poll Question 1:

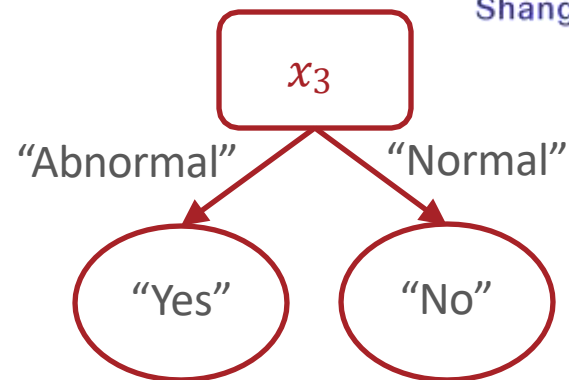
Which of the following did you bring to class today?
Select all that apply

- A. A smartphone
- B. A computer
- C. A smart watch
- D. No device



Recall: Decision Stump

x_1 Family History	x_2 Resting Blood Pressure	x_3 Cholesterol	y Heart Disease?
Yes	Low	Normal	No
No	Medium	Normal	No
No	Low	Abnormal	Yes
Yes	Medium	Normal	Yes
Yes	High	Abnormal	Yes



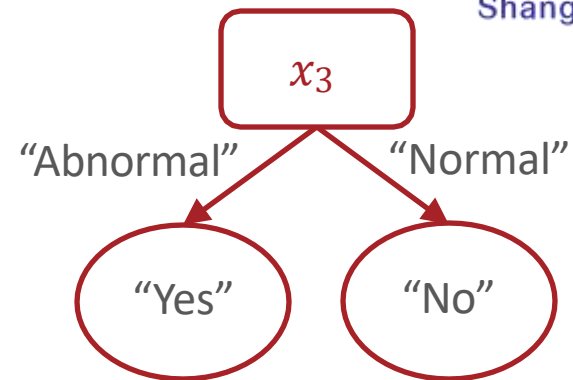
Recall: Decision Stump Questions

1. How can we pick which feature to split on?
2. Why stop at just one feature? **Don't!**
 - a) If we split on more than one feature, how do we decide the order to split on?

From Decision Stump

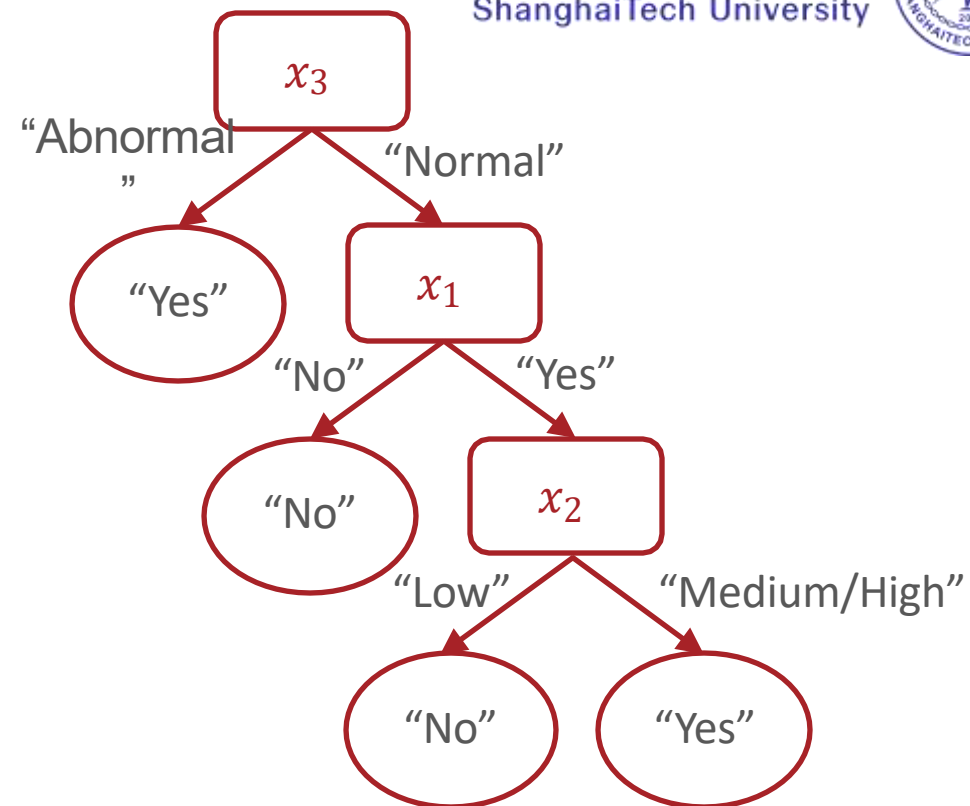
...

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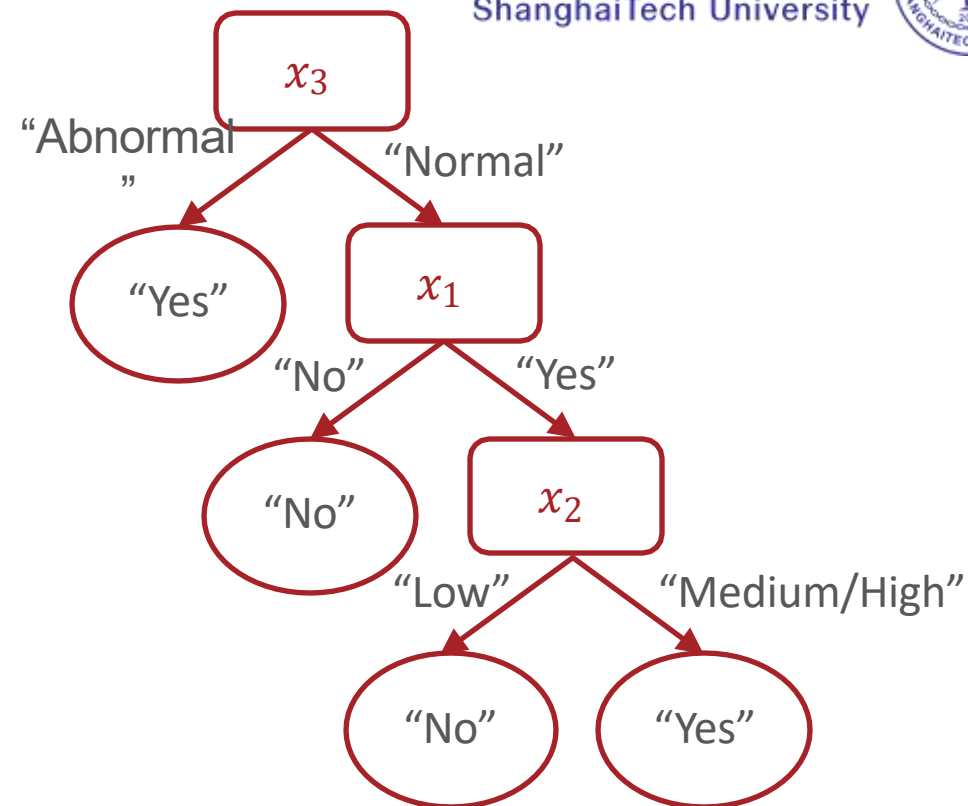
From Decision Stump to Decision Tree

x_1 Family History	x_2 Resting Blood Pressure	x_3 Cholesterol	y Heart Disease?
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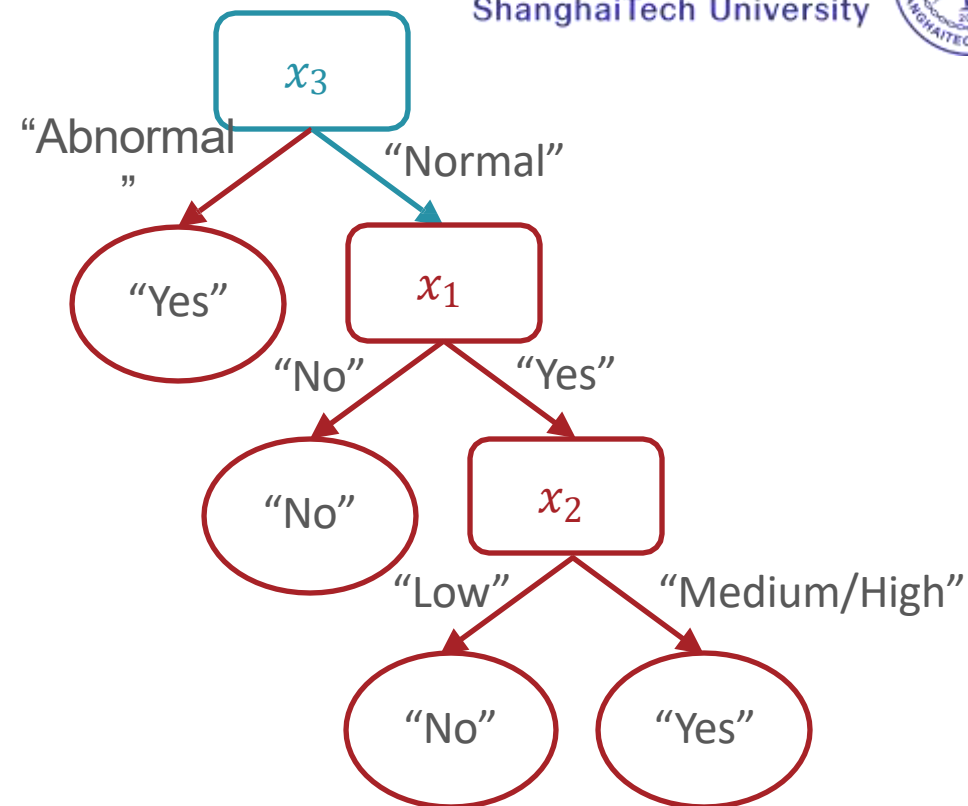
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No	Medium	Normal	No
No	Low	Abnormal	Yes
Yes	Medium	Normal	Yes
Yes	High	Abnormal	Yes
No	High	Normal	No]



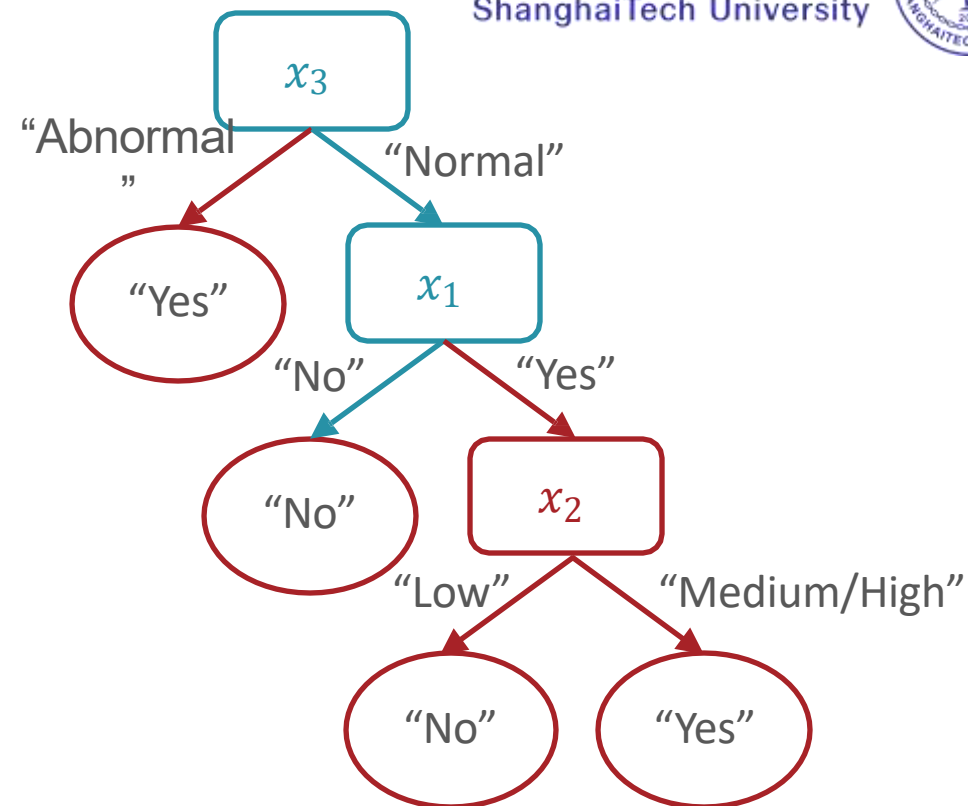
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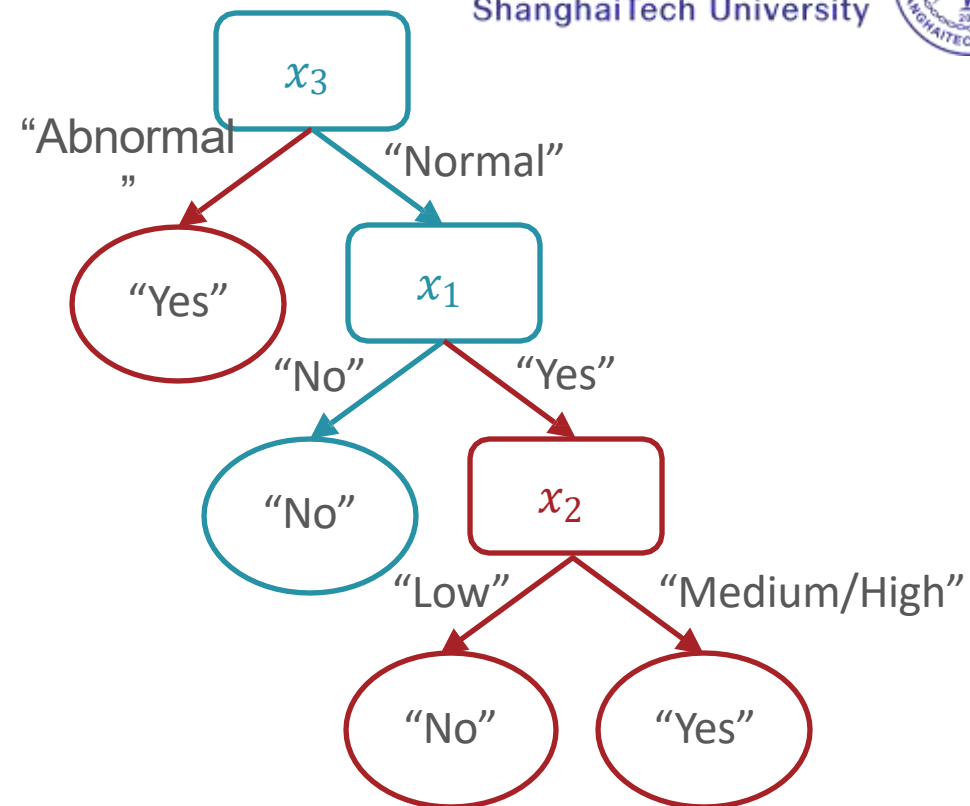
From Decision Stump to Decision Tree

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From Decision Stump to Decision Tree

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Yes	Medium	Normal	Yes
Yes	High	Abnormal	Yes
No	High	Normal	No



Decision Tree: Pseudocode



```
def  $h(x')$ :
```

```
- walk from root node to a leaf node
```

```
while(true):
```

```
    if current node is internal (non-leaf):
```

```
        check the associated attribute,  $x_d$ 
```

```
        go down branch according to  $x'_d$ 
```

```
    if current node is a leaf node:
```

```
        return label stored at that leaf
```

Decision Tree Questions



1. How can we pick which feature to split on?
2. Why stop at just one feature?
 - a) If we split on more than one feature, how do we decide the order to split on?

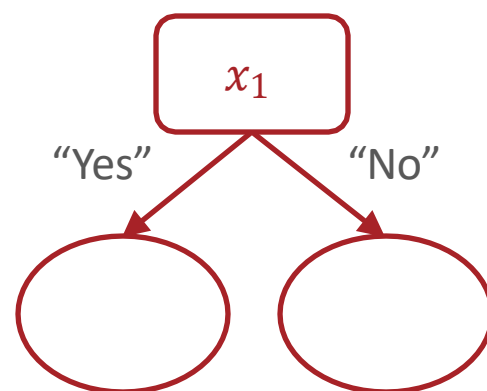
Splitting Criterion



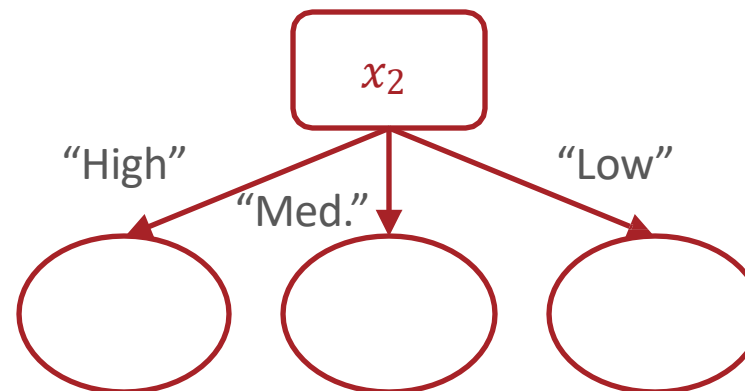
- A **splitting criterion** is a function that measures how good or useful splitting on a particular feature is *for a specified dataset*
- Idea: when deciding which feature to split on, use the one that optimizes the splitting criterion

Training Error Rate as a Splitting Criterion

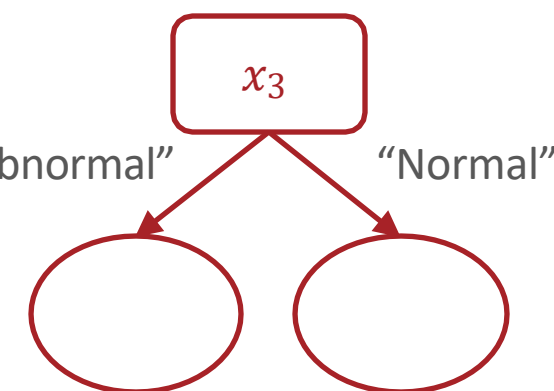
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Yes	Low	Normal	No
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Yes	Medium	Normal	Yes
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Training error rate:



Training error rate:



Training error rate:

Poll Question 2:

■ Which feature would you split on using training error rate as the splitting criterion?

x_1	x_2	y
1	0	0
1	0	0
1	0	1
1	0	1
1	1	1
1	1	1
1	1	1
1	1	1

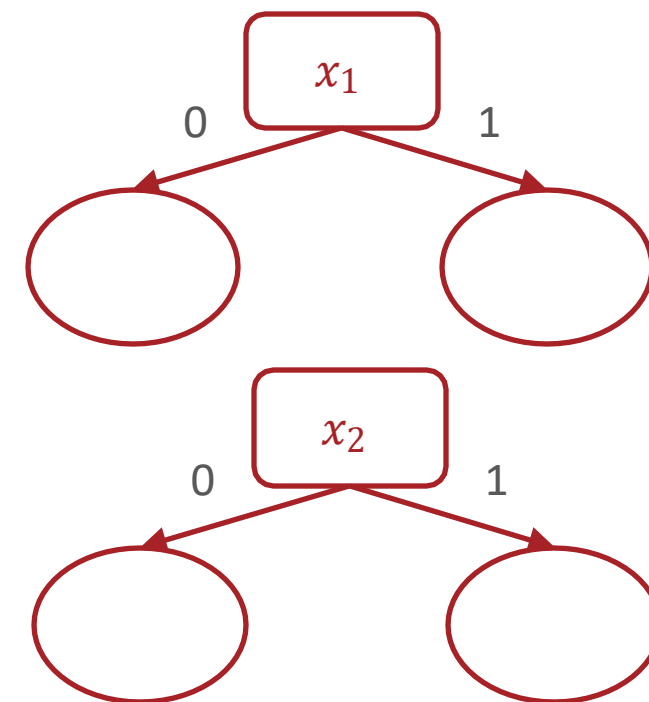


- A. x_1
- B. x_2
- C. Either x_1 or x_2
- D. Neither x_1 nor x_2

Poll Question 2:

■ Which feature would you split on using training error rate as the splitting criterion?

x_1	x_2	y
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1	0	0
1	0	1
1	0	1
1	1	1
1	1	1
1	1	1
1	1	1



Training error rate:

Splitting Criterion



- A **splitting criterion** is a function that measures how good or useful splitting on a particular feature is *for a specified dataset*
- Idea: when deciding which feature to split on, use the one that optimizes the splitting criterion
- Potential splitting criteria:
 - Training error rate (minimize)
 - Gini impurity (minimize) → CART algorithm
 - Mutual information (maximize) → ID3 algorithm

Splitting Criterion



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- Idea: when deciding which feature to split on, use the one that optimizes the splitting criterion
- Potential splitting criteria:
 - Training error rate (minimize)
 - Gini impurity (minimize) → CART algorithm
 - **Mutual information** (maximize) → ID3 algorithm

Entropy



- The **entropy** of a *random variable* describes the uncertainty of its outcome: the higher the entropy, the less certain we are about what the outcome will be.

$$H(X) = - \sum_{v \in V(X)} P(X = v) \log_2(P(X = v))$$

where X is a (discrete) random variable

$V(X)$ is the set of possible values X can take on

Entropy



- The **entropy** of a *set* describes how uniform or pure it is: the higher the entropy, the more impure or “mixed-up” the set is

$$H(S) = - \sum_{v \in V(S)} \frac{|S_v|}{|S|} \log_2 \left(\frac{|S_v|}{|S|} \right)$$

where S is a collection of values,

$V(S)$ is the set of unique values in S

S_v is the collection of elements in S with value v

- If all the elements in S are the same, then

$$H(S) = -1 \log_2 (1) = 0$$

Entropy

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where S is a collection of values,

$V(S)$ is the set of unique values in S

S_v is the collection of elements in S with value v

- If S is split fifty-fifty between two values, then

$$H(S) = -\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) = -\log_2 \left(\frac{1}{2} \right) = 1$$

Mutual Information



- The **mutual information** between *two random variables* describes how much clarity knowing the value of one random variables provides about the other

$$I(Y; X) = H(Y) - H(Y|X)$$

$$= H(Y) - \sum_{v \in V(X)} P(X = v) H(Y|X = v)$$

where X and Y are random variables

$V(X)$ is the set of possible values X can take on

$H(Y|X = v)$ is the conditional entropy of Y given $X = v$

Mutual Information



- The **mutual information** between *a feature and the label* describes how much clarity knowing the feature provides about the label

$$\begin{aligned} I(y; x_d) &= H(y) - H(y|x_d) \\ &= H(y) - \sum_{v \in V(x_d)} f_v \left(H(Y_{x_d=v}) \right) \end{aligned}$$

where x_d is a feature and y is the set of all labels

$V(x_d)$ is the set of possible values x_d can take on

f_v is the fraction of data points where $x_d = v$

$Y_{x_d=v}$ is the set of all labels where $x_d = v$

Mutual Information Example

x_d	y
1	1
1	1
0	0
0	0

$$\begin{aligned}
 I(x_d, Y) &= H(Y) - \sum_{v \in V(x_d)} (f_v) \left(H(Y_{x_d=v}) \right) \\
 &= 1 - \frac{1}{2} H(Y_{x_d=0}) - \frac{1}{2} H(Y_{x_d=1}) \\
 &= 1 - \frac{1}{2} (0) - \frac{1}{2} (0) = 1
 \end{aligned}$$

Mutual Information Example

x_d	y
1	1
0	1
1	0
0	0

$$\begin{aligned}
 I(x_d, Y) &= H(Y) - \sum_{v \in V(x_d)} (f_v) \left(H(Y_{x_d=v}) \right) \\
 &= 1 - \frac{1}{2} H(Y_{x_d=0}) - \frac{1}{2} H(Y_{x_d=1}) \\
 &= 1 - \frac{1}{2} (1) - \frac{1}{2} (1) = 0
 \end{aligned}$$

Poll Question 3:

Which feature would you split on using mutual information as the splitting criterion?

x_1	x_2	y
1	0	0
1	0	0
1	0	1
1	0	1
1	1	1
1	1	1
1	1	1
1	1	1

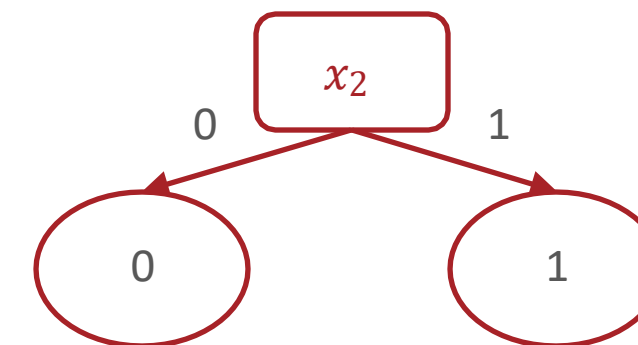
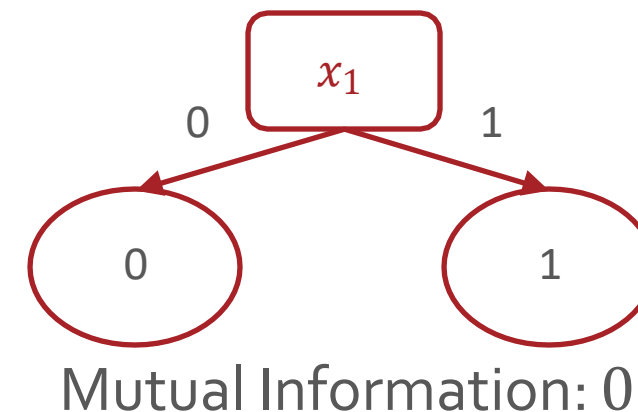


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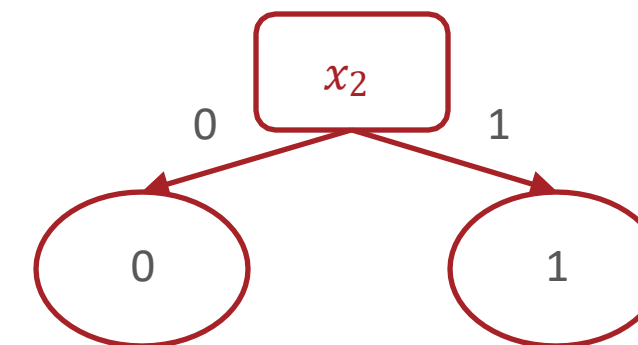
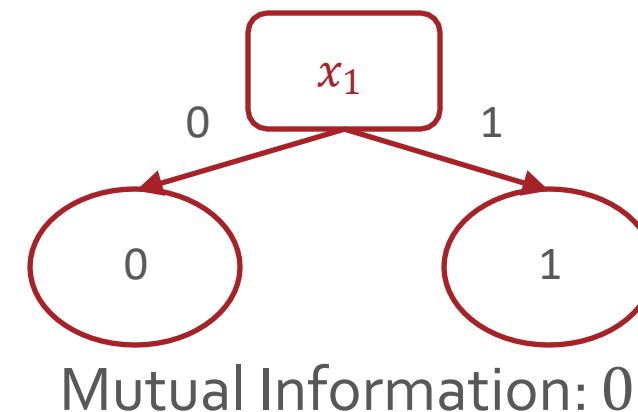


Mutual Information: $H(Y) - \frac{1}{2}H(Y_{x_2=0}) - \frac{1}{2}H(Y_{x_2=1})$

Poll Question 3:

Which feature would you split on using mutual information as the splitting criterion?

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1	0	0
1	0	0
1	0	1
1	0	1
1	1	1
1	1	1
1	1	1
1	1	1



$$\text{Mutual Information: } -\frac{2}{8}\log_2\frac{2}{8} - \frac{2}{8}\log_2\frac{2}{8} - \frac{6}{8}\log_2\frac{6}{8} - \frac{1}{2}(1) - \frac{1}{2}(0) \approx 0.31$$

Decision Tree: Questions



1. How can we pick which feature to split on?
2. Why stop at just one feature?
 - a) If we split on more than one feature, how do we decide the order to split on?

Decision Tree: Pseudocode

```
def train( $\mathcal{D}$ ):  
    store root = tree_recurse( $\mathcal{D}$ )  
  
def tree_recurse( $\mathcal{D}'$ ):  
    q = new node()  
    base case – if (SOME CONDITION):  
    recursion – else:  
        find best attribute to split on,  $x_d$   
        q.split =  $x_d$   
        for  $v$  in  $V(x_d)$ , all possible values of  $x_d$ :  
             $\mathcal{D}_v = \{(x^{(n)}, y^{(n)}) \in \mathcal{D} \mid x_d^{(n)} = v\}$   
            q.children( $v$ ) = tree_recurse( $\mathcal{D}_v$ )  
    return q
```

Decision Tree: Pseudocode

```
def train( $\mathcal{D}$ ):  
    store root = tree_recurse( $\mathcal{D}$ )  
def tree_recurse( $\mathcal{D}'$ ):  
    q = new node()  
    base case – if ( $\mathcal{D}'$  is empty OR  
        all labels in  $\mathcal{D}'$  are the same OR  
        all features in  $\mathcal{D}'$  are identical OR  
        some other stopping criterion):  
        q.label = majority_vote( $\mathcal{D}'$ )  
  
    recursion – else:  
        return q
```

Q & A:

Wait, how could we end up calling `tree_recurse` with an empty dataset in the first place?

- Given some subset of our dataset, it could be the case that we choose to split on some feature where not every value that the feature could take on appears in the subset
 - This could happen if we know something about the feature *a priori* or we observe that the feature takes on more values in a different subset/the entire training dataset.
- In this case we would still want to make a branch for it in our decision tree because at inference time, some new data point might come along that goes down that branch

Q & A:

Okay, so what should we predict in leaf nodes with no training data?

- Well, there isn't really a majority label, so we could return a random label or a majority vote over the entire training dataset.
- This is related to the question of “what should we predict if some feature in our test dataset takes on a value we didn't observe in the training dataset?”
 - Going down a branch corresponding to an unseen feature value is like hitting a leaf node with no training data.

Bluntine & Niblett (1992) Splitting Criteria Experiment s

Table 1. Properties of the data set

Data Set	Classes	Attr.s	Training Set	Test Set
hypo	4	29	1000	2772
breast	2	9	200	86
tumor	22	18	237	102
lymph	4	18	103	45
LED	10	7	200	1800
mush	2	22	200	7924
votes	2	17	200	235
votes1	2	16	200	235
iris	3	4	100	50
glass	7	9	100	114
xd6	2	10	200	400
pole	2	4	200	1647

Bluntine & Niblett (1992) Splitting Criteria Experiments

Table 3. Error for different splitting rules (pruned trees).

Data Set	Splitting Rule		
	GINI	Info. Gain	Random
hypo	1.01 $\pm$ 0.29	0.95 $\pm$ 0.22	7.44 $\pm$ 0.53
breast	28.66 $\pm$ 3.87	28.49 $\pm$ 4.28	29.65 $\pm$ 4.97
tumor	60.88 $\pm$ 5.44	62.70 $\pm$ 3.89	67.94 $\pm$ 5.68
lymph	24.44 $\pm$ 6.92	24.00 $\pm$ 6.87	32.33 $\pm$ 11.25
LED	33.77 $\pm$ 3.06	32.89 $\pm$ 2.59	38.18 $\pm$ 4.57
mush	1.44 $\pm$ 0.47	1.44 $\pm$ 0.47	8.77 $\pm$ 4.65
votes	4.47 $\pm$ 0.95	4.57 $\pm$ 0.87	12.40 $\pm$ 4.56
votes1	12.79 $\pm$ 1.48	13.04 $\pm$ 1.65	15.62 $\pm$ 2.73
iris	5.00 $\pm$ 3.08	4.90 $\pm$ 3.08	14.20 $\pm$ 6.77
glass	39.56 $\pm$ 6.20	50.57 $\pm$ 6.73	53.20 $\pm$ 5.01
xd6	22.14 $\pm$ 3.23	22.17 $\pm$ 3.36	31.86 $\pm$ 3.62
pole	15.43 $\pm$ 1.51	15.47 $\pm$ 0.88	26.38 $\pm$ 6.92

Key takeaway: Gini impurity and Information gain (aka mutual information) are nearly identical

Bluntine & Niblett (1992) Splitting Criteria Experiment S

Table 4. Difference and significance of error for GINI splitting rule versus others.

Data Set	Splitting Rule	
	Info. Gain	Random
hypo	-0.06 (0.82)	6.43 (1.00)
breast	-0.17 (0.23)	0.99 (0.72)
tumor	1.81 (0.84)	7.06 (0.99)
lymph	-0.44 (0.83)	7.89 (0.99)
LED	0.12 (0.17)	5.41 (0.99)
mush	0.00 (0.00)	7.32 (0.99)
votes	0.11 (0.55)	7.94 (0.99)
votes1	0.26 (0.47)	2.83 (0.99)
iris	-0.10 (0.67)	9.20 (0.99)
glass	1.01 (0.50)	13.64 (0.99)
xd6	0.04 (0.11)	9.72 (0.99)
pole	0.03 (0.11)	10.95 (0.99)

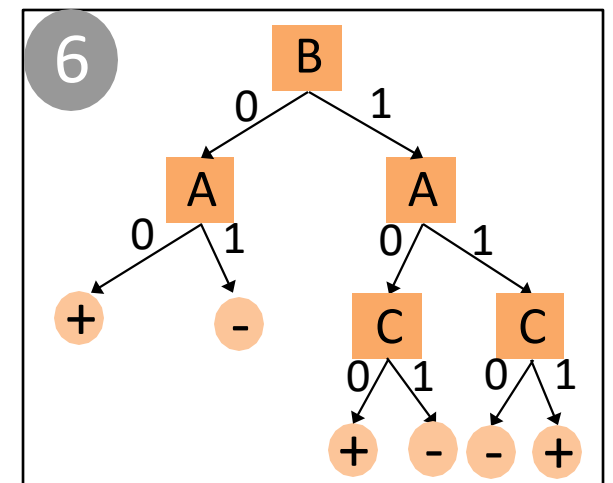
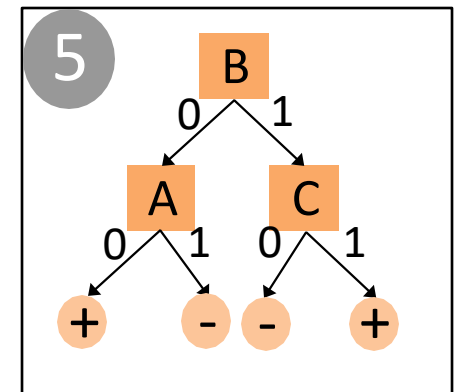
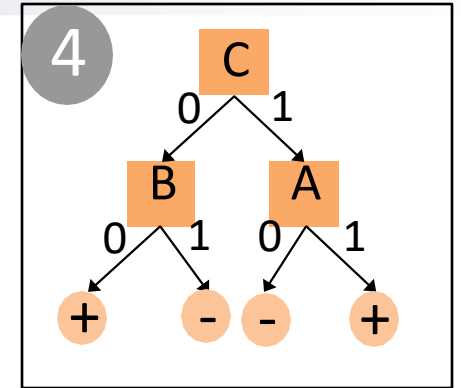
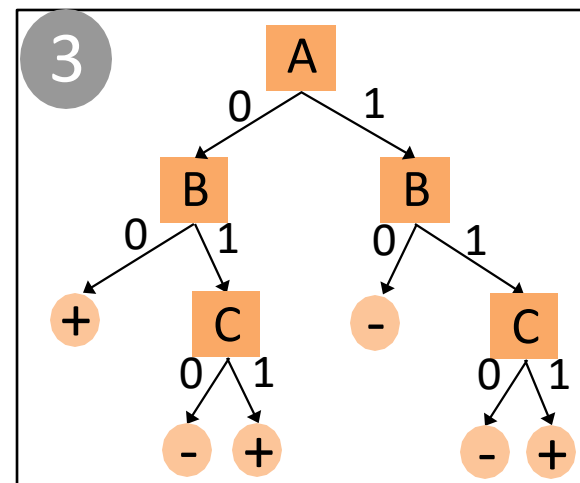
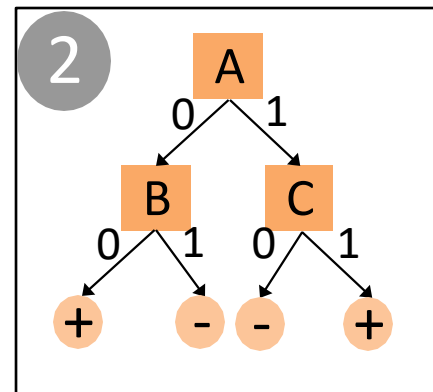
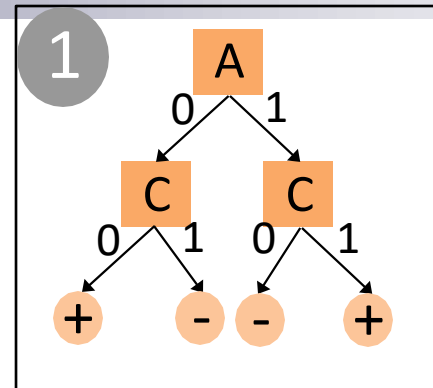
average difference
in error between
the splitting
criteria

statistical significance of
the difference according to
a two-sided paired t-test

Source: <https://link.springer.com/content/pdf/10.1023/A:1022686419106.pdf>

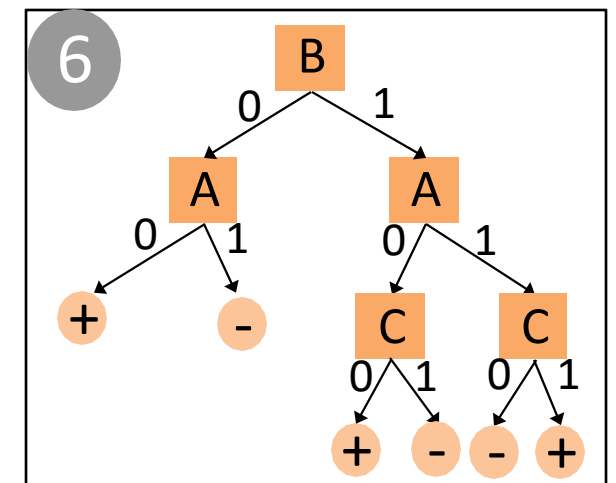
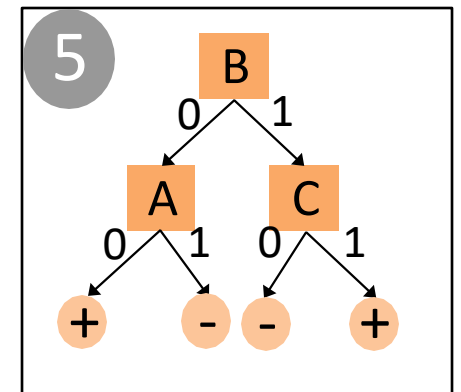
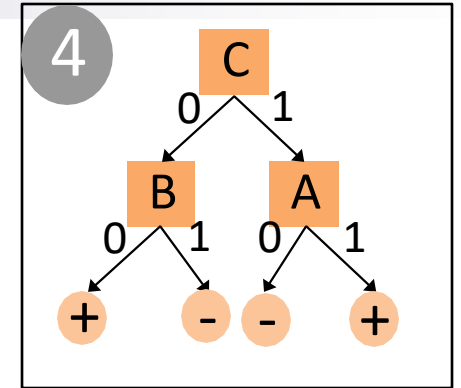
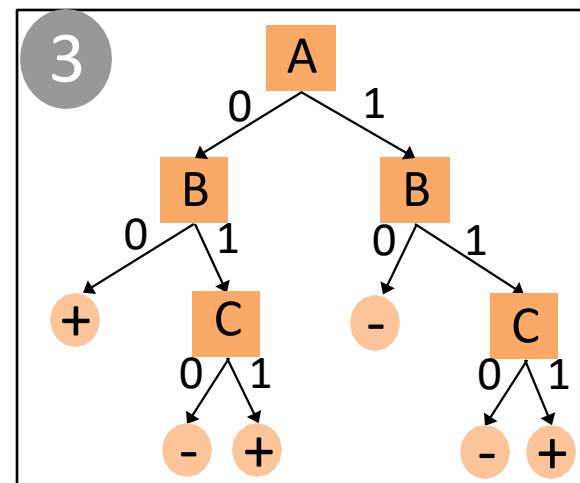
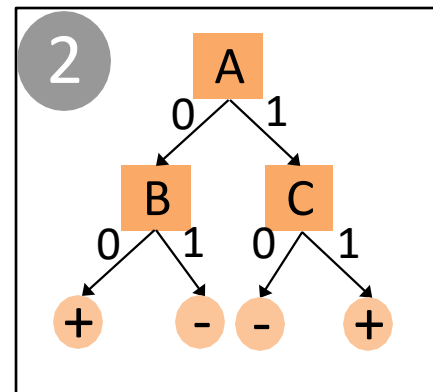
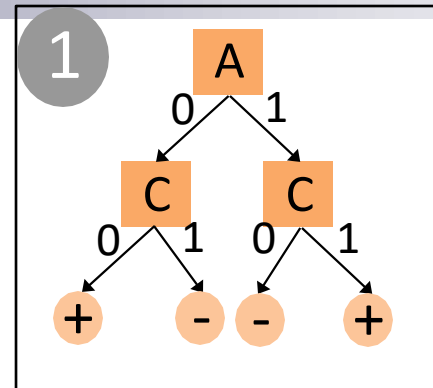
Poll Question 1:
Which decision tree would you learn if you used training error rate as the splitting criterion? Break ties alphabetically.

<i>A</i>	<i>B</i>	<i>C</i>	<i>y</i>
0	0	0	+
0	0	1	+
0	1	0	-
0	1	1	+
1	0	0	-
1	0	1	-
1	1	0	-
1	1	1	+



Poll Question 2:
Which decision tree is the smallest decision tree that achieves the lowest possible training error?

<i>A</i>	<i>B</i>	<i>C</i>	<i>y</i>
0	0	0	+
0	0	1	+
0	1	0	-
0	1	1	+
1	0	0	-
1	0	1	-
1	1	0	-
1	1	1	+



Decision Trees: Inductive Bias



- The **inductive bias** of a machine learning algorithm is the principle by which it generalizes to unseen examples
- What is the inductive bias of the ID3 algorithm?
 - Try to find the **_smallest decision_** tree that achieves a **_low/zero training error_** with **_high mutual information_** features at the top
- Occam's razor: try to find the “simplest” (e.g., smallest decision tree) classifier that explains the training dataset

Decision Trees: Pros & Cons

- Pros
 - Interpretable
 - Efficient (computational cost and storage)
 - Can be used for classification and regression tasks
 - Compatible with categorical and real-valued features
- Cons
 - Learned greedily: each split only considers the immediate impact on the splitting criterion
 - Not guaranteed to find the smallest (fewest number of splits) tree that achieves a training error rate of 0.

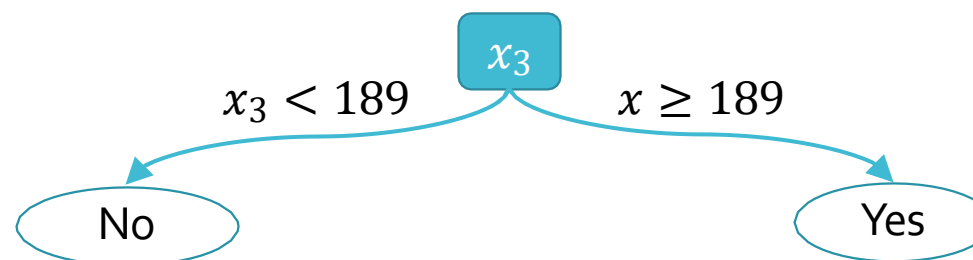
Real-Valued Features: Example

x_3 Cholesterol	y Heart Disease?
174	No
155	No
163	Yes
233	Yes
197	Yes
181	No
244	Yes
245	No
178	No
221	Yes



x_3 Cholesterol	y Heart Disease?
155	No
163	Yes
174	No
178	No
181	No
197	Yes
221	Yes
233	Yes
244	Yes
245	No

← $x_3 < 189$



Real-Valued Features: Example

x_3 Cholesterol	y Heart Disease?
174	No
155	No
163	Yes
233	Yes
197	Yes
181	No
244	Yes
245	No
178	No
221	Yes

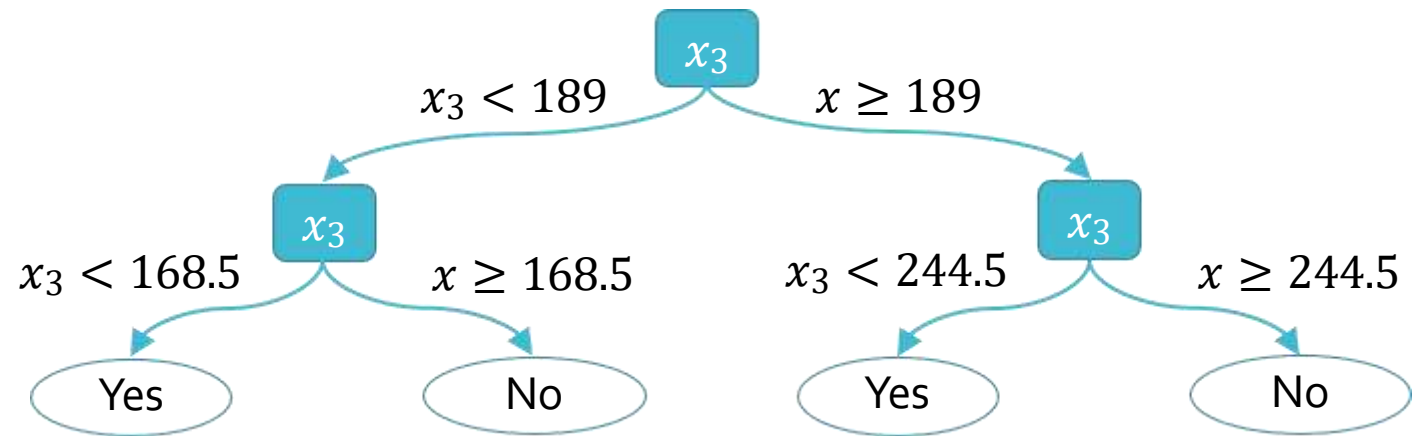


x_3 Cholesterol	y Heart Disease?
155	No
163	Yes
174	No
178	No
181	No
197	Yes
221	Yes
233	Yes
244	Yes
245	No

← $x_3 < 168.5$

← $x_3 < 189$

← $x_3 < 244.5$



Decision Trees: Pros & Cons



- Pros
 - Interpretable
 - Efficient (computational cost and storage)
 - Can be used for classification and regression tasks
 - Compatible with categorical and real-valued features
- Cons
 - Learned greedily: each split only considers the immediate impact on the splitting criterion
 - Not guaranteed to find the smallest (fewest number of splits) tree that achieves a training error rate of 0.
 - Liable to overfit!

Overfitting

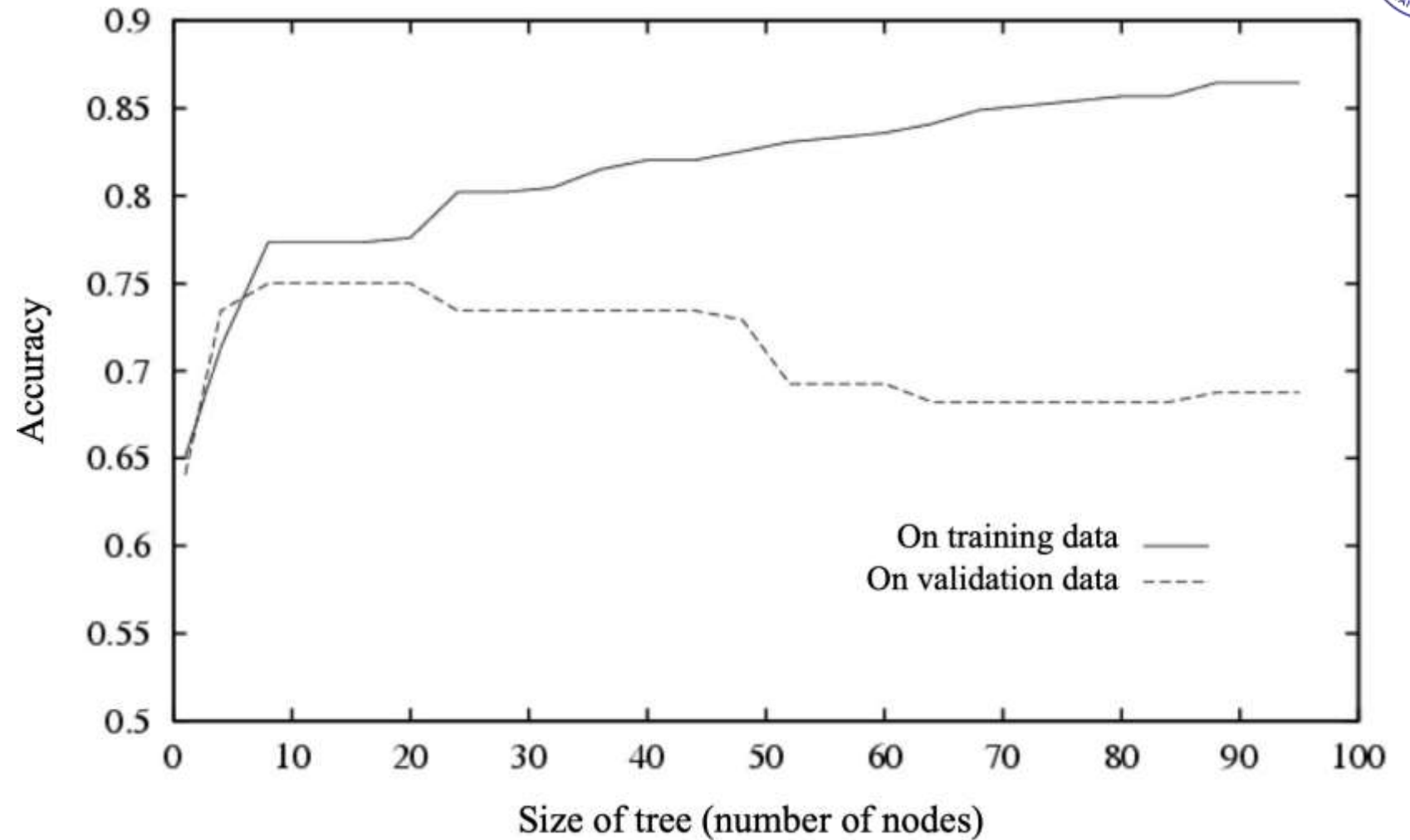
- Overfitting occurs when the classifier (or model)...
 - is too complex
 - fits noise or “outliers” in the training dataset as opposed to the actual pattern of interest
 - doesn’t have enough inductive bias pushing it to generalize e.g., the memorizer
- Underfitting occurs when the classifier (or model)...
 - is too simple
 - can’t capture the actual pattern of interest in the training dataset
 - has too much inductive bias e.g., the majority vote classifier

Recall: Different Kinds of Error



- Training error rate = $err(h, \mathcal{D}_{train})$
- Test error rate = $err(h, \mathcal{D}_{test})$
- True error rate = $err(h)$
 - = the error rate of h on all possible examples
 - In machine learning, this is the quantity that we care about but, in most cases, it is unknowable.
- Overfitting occurs when $err(h) > err(h, \mathcal{D}_{train})$
 - $err(h) - err(h, \mathcal{D}_{train})$ can be thought of as a measure of overfitting

Overfitting in Decision Trees



Combating Overfitting in Decision Trees

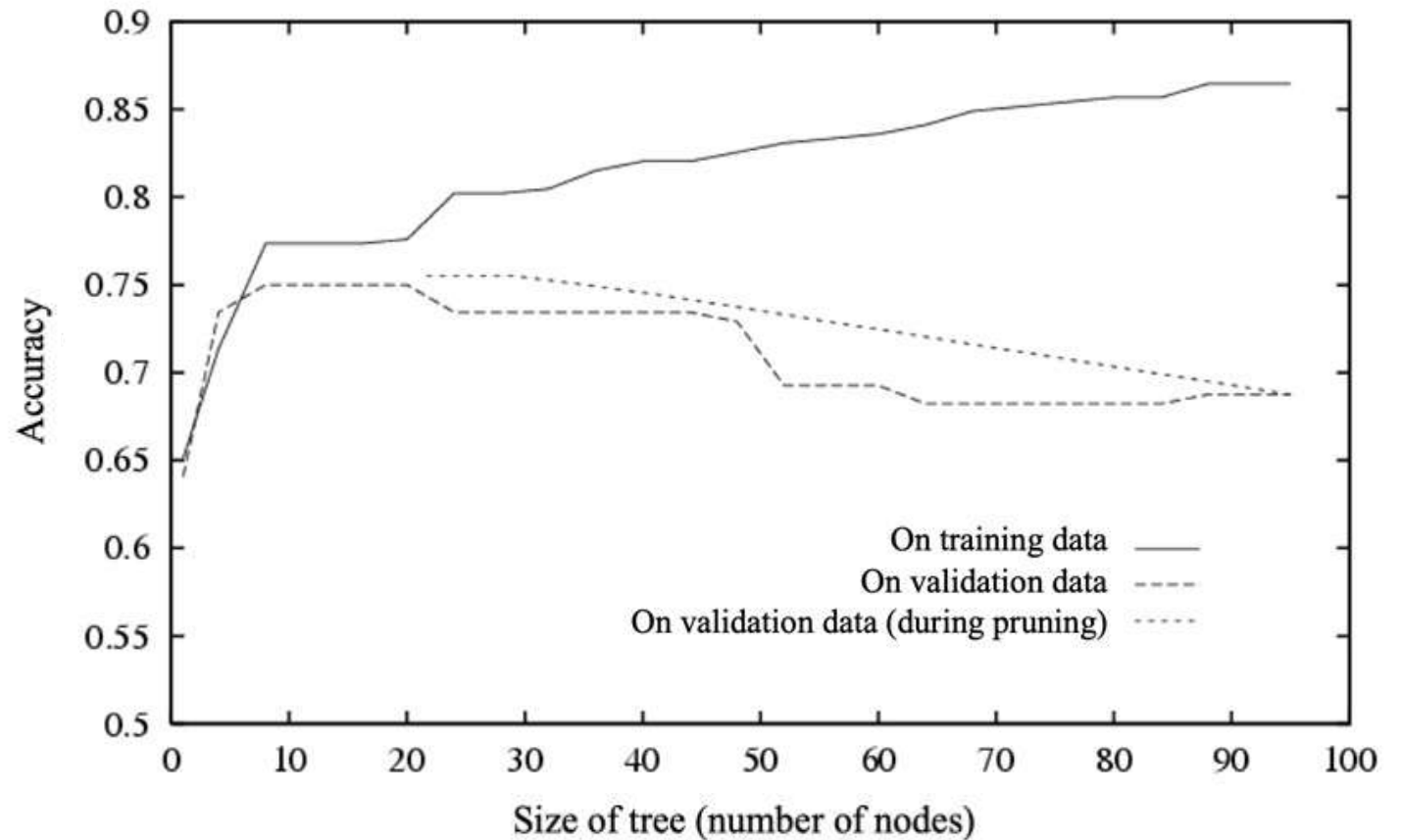


- Heuristics:
 - Do not split leaves past a fixed depth, δ
 - Do not split leaves with fewer than c data points
 - Do not split leaves where the maximal information gain is less than τ
- Take a majority vote in impure leaves

Combating Overfitting in Decision Trees



1. Split data in two: **training** dataset and **validation** dataset
2. Grow the full tree using the **training** dataset
3. Repeatedly prune the tree:
 - Evaluate each split using a **validation** dataset by comparing the validation error rate **with and without** that split
 - (Greedy) remove the split that most decreases the validation error rate
 - Stop if no split improves validation error, otherwise repeat



Decision Tree Learning Objectives

You should be able to...

1. Implement decision tree training and prediction
2. Use effective splitting criteria for decision trees and be able to define entropy, conditional entropy, and mutual information / information gain
3. Explain the difference between memorization and generalization [CIML]
4. Describe the inductive bias of a decision tree
5. Formalize a learning problem by identifying the input space, output space, hypothesis space, and target function
6. Explain the difference between true error and training error
7. Judge whether a decision tree is "underfitting" or "overfitting"
8. Implement a pruning or early stopping method to combat overfitting in decision tree learning